Portfolio Technical Architecture & Implementation Guide

PYTHON FULL-STACK & MODERN WEB APPLICATION DOCUMENTATION

Engineer: Tripurosundaro Behera
Education: MCA (Post Graduate, CGPA 8.47/10)
Domain: tripurosundaro.me
Repository: TRIPUROSUNDAROOBEHERA/PortfolioWebsite

✓ Live Status: Deployed (HTTP 200 OK) | React 18 + Vite 5 | Tailwind CSS 3 | Python 3.12 + Django REST Framework | Test Suite: 23 Suites / 61 Tests (100% Passed)

1 System Architecture Overview

The portfolio is engineered as a decoupled, modern multi-tier web application designed for high performance, zero-lag responsiveness, and seamless dual deployment across edge hosting and dedicated backend environments.

Client & Presentation Tier

- **Framework:** React 18 with Vite 5 compilation engine.
- **Routing:** React Router v6 with dynamic `basename` routing supporting both subpath GitHub Pages and apex custom domains.
- **Styling:** Tailwind CSS utility layer with mobile-first responsive breakpoints (xs: 375px, sm: 640px, md: 768px, lg: 1024px, xl: 1280px).
- **Analytics Visualization:** Recharts charting suite powering IDS ML benchmarks and data evaluation matrices.

Server & API Services Tier

- **Engine:** Python 3.12+ with Django 5.x and Django REST Framework.
- **Architecture:** Modular app hierarchy (`profile`, `projects`, `skills`, `education`, `contact`, `dashboard`).
- **Data Persistence:** Dual database support (SQLite for local micro-services; PostgreSQL/MySQL for production deployments).
- **Security:** Django CORS headers, CSRF token validation, request rate-limiting, and XSS sanitization.

2 Frontend Deep-Dive & Zero-Lag Engineering

Module / Component	Implementation Details	Performance & User Experience Highlights
Adaptive Hero Section	Dual-layer render: 3D interactive flip on desktop; glowing blue circular border avatar on mobile.	Hardware-accelerated CSS transforms; 0% cumulative layout shift (CLS).
A4 Verified Resume Modal	Full-length A4 document container with symmetric margins, custom gray scrollbar, print & PDF export.	Verified MCA CGPA (8.47/10), projects, and direct links to LinkedIn, GitHub, and Portfolio.
TripGen Itinerary Showcase	Split-screen banner with staggered waterfall itinerary schedule, security modal, and testing logs.	Dynamic destination timeline with live day-by-day activity tabs and modal overlays.
IDS Cyber Threat Detector	NSL-KDD benchmark evaluation panel with 99.0% XGBoost accuracy and Random Forest confusion matrix.	Interactive Recharts data plots with hover tooltips and attack classification distribution.
English Master Pro App	Cross-platform Flutter showcase featuring mobile carousel, feature pillars, and generated APK download QR code.	Smooth touch gestures, auto-advancing carousel, and integrated preview video container.

Zero-Lag Performance Optimizations

- **Vite Manual Chunk Splitting:** Extracted `vendor-react` (159 kB), `vendor-charts`, and `vendor-icons` into isolated chunks, eliminating bundle bottleneck warnings.
- **Passive Event Listeners:** All scroll observers in `Home.jsx` utilize `{ passive: true }`, ensuring unblocked UI threads and constant 60 FPS scrolling.
- **GPU Compositing:** Key transitions leverage `transform: translate3d(0, 0, 0)` and `will-change: transform` to bypass CPU repaint cycles.

Deployment Pipeline & Operations Manual

CI/CD, DNS BINDING, AND MAINTENANCE REFERENCE

Live URL: https://tripurosundaroobehera.github.io/PortfolioWebsite/
Custom Domain: https://tripurosundaro.me
Build Tool: GitHub Actions (actions/deploy-pages@v4)

3 Automated CI/CD & GitHub Pages Edge Deployment

Continuous Deployment is managed through an automated GitHub Actions workflow (`.github/workflows/deploy-pages.yml`). Every git push to `main` initiates a fresh, verified production deployment.

```
name: Deploy Portfolio to GitHub Pages on: { push: { branches: ["main"] } } permissions: { contents: read, pages: write, id-token: write } jobs: build-and-deploy: runs-on: ubuntu-latest steps: - uses: actions/checkout@v4 - uses: actions/setup-node@v4 with: { node-version: 20 } - run: cd frontend && npm ci && npm run build - run: cp frontend/dist/index.html frontend/dist/404.html # SPA Route Redirect - uses: actions/upload-pages-artifact@v3 with: { path: './frontend/dist' } - uses: actions/deploy-pages@v4
```

Dual-Host Router Compatibility

Configured in `frontend/src/main.jsx` to seamlessly detect URL environment:

```
const basename =
window.location.pathname.startsWith('/PortfolioWebsite')
? '/PortfolioWebsite' : '';
```

Client SPA Fallback (404.html)

GitHub Pages serves static files. When visitors refresh subroutes (e.g. `/projects`), the edge CDN serves `404.html` (an exact copy of `index.html`), enabling React Router to hydrate and maintain clean navigation without 404 errors.

4 Custom Domain DNS Binding (tripurosundaro.me via Namecheap)

To point the domain acquired through the GitHub Student Developer Pack to your GitHub Pages build, add these records in **Namecheap Advanced DNS**:

Record Type	Host	Destination Value	TTL	Purpose
A Record	`@`	`185.199.108.153`	Automatic (1 min)	GitHub Pages Anycast Load Balancer Node 1
A Record	`@`	`185.199.109.153`	Automatic (1 min)	GitHub Pages Anycast Load Balancer Node 2
A Record	`@`	`185.199.110.153`	Automatic (1 min)	GitHub Pages Anycast Load Balancer Node 3
A Record	`@`	`185.199.111.153`	Automatic (1 min)	GitHub Pages Anycast Load Balancer Node 4
CNAME Record	`www`	`TRIPUROSUNDAROOBEHERA.github.io.`	Automatic (1 min)	Subdomain Alias Routing

5 Local Developer Operations Reference

1. Running Local Frontend Server

```
cd frontend npm run dev # Live preview at
http://localhost:5173
```

2. Running Test Verification Suite

```
cd frontend npm run test:run # Executes all 61
tests across 23 test suites
```

> **Publishing Future Updates:** Any commit pushed via `git push origin main` or by running `publish_to_github.bat` will automatically re-build and publish your website across the global CDN within ~45 seconds.